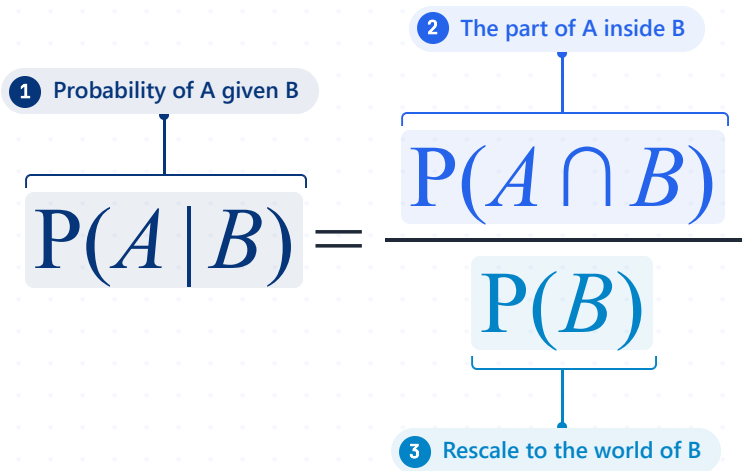
THE FORMULA, PART BY PART

Conditional probability

one rule: keep the part of A that lies inside B, then rescale by the size of B



1 Probability of A given B $P(A \mid B)$

The bar reads "given": everything to its right is the restricted world we now work in. Example: two dice, A = first die even, B = sum 8. Of the 5 outcomes with sum 8, 3 have an even first die, so P(A | B) = 3/5.

2 The part of A inside B $P(A \cap B)$

The rest of A lies outside the condition and is dropped. Measured against the whole sample space, this is not yet the answer. Example: 3 of the 36 outcomes have an even first die and sum 8, so P(A ∩ B) = 3/36.

3 Rescale to the world of B $P(B)$

Dividing by P(B) turns B into the whole sample space. It is a rescaling, not a new law of probability. Example: P(sum 8) = 5/36, and (3/36) ÷ (5/36) = 3/5: each of the 5 remaining outcomes now carries 1/5.